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Advances in blood biomarkers for Alzheimer disease (AD): A review

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Abstract

Alzheimer disease (AD) and Alzheimer Disease and Related Dementias (AD/ADRD) are growing public health challenges globally affecting millions of older adults, necessitating concerted efforts to advance our understanding and management of these conditions. AD is a progressive neurodegenerative disorder characterized pathologically by amyloid plaques and tau neurofibrillary tangles that are the primary cause of dementia in older individuals. Early and accurate diagnosis of AD dementia is crucial for effective intervention and treatment but has proven challenging to accomplish. Although testing for AD brain pathology with cerebrospinal fluid (CSF) or positron emission tomography (PET) has been available for over 2 decades, most patients never underwent this testing because of inaccessibility, high out-of-pocket costs, perceived risks, and the lack of AD-specific treatments. However, in recent years, rapid progress has been made in developing blood biomarkers for AD/ADRD. Consequently, blood biomarkers have emerged as promising tools for non-invasive and cost-effective diagnosis, prognosis, and monitoring of AD progression. This review presents the evolving landscape of blood biomarkers in AD/ADRD and explores their potential applications in clinical practice for early detection, prognosis, and therapeutic interventions. It covers recent advances in blood biomarkers, including amyloid beta (A β) peptides, tau protein, neurofilament light chain (NfL), and glial fibrillary acidic protein (GFAP). It also discusses their diagnostic and prognostic utility while addressing associated challenges and limitations. Future research directions in this rapidly evolving field are also proposed.

KEYWORDS

Alzheimer disease, blood test, blood-based biomarkers, diagnosis, plasma biomarkers

1 | INTRODUCTION

Alzheimer disease (AD) is the most common cause of dementia, affecting 55 million older adults worldwide and presenting a significant burden on healthcare systems.¹ AD and Alzheimer Disease and Related Dementias (AD/ADRD) encompass the neurodegenerative

conditions that are characterized by progressive cognitive decline, functional impairment, and behavioral changes.^{2–6} The number of people with dementia is projected to reach 78 million in 2030 and 139 million in 2050, representing 10 million new cases of dementia each year worldwide.¹ With this significant increase in dementia prevalence as populations age, addressing AD/ADRD has become a

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priority within public health agendas globally. The increasing prevalence of ADRD presents substantial socioeconomic and healthcare burdens, affecting individuals, families, and healthcare systems alike. Significantly, ADRD not only affects cognitive functions but also leads to functional decline, loss of independence, and increased caregiving needs. Furthermore, ADRD is associated with significant healthcare costs, including medical care, long-term care, and caregiving expenses.⁷ The current global annual cost estimation for dementia exceeds 1.3 trillion USD, with projections indicating a rise to 2.8 trillion USD by 2030.¹

AD is defined by its pathology rather than its symptoms. The pathological characteristics of AD were first identified by Alois Alzheimer in 1906.⁸ They include the presence of extracellular amyloid plaques comprised of amyloid- β peptide and intracellular neurofibrillary tangles comprised of the microtubule-associated protein tau.^{5,9} Amyloid plaques and tau tangles are often associated with synaptic dysfunction, synaptic loss, and neurodegeneration. Biomarkers that reflect amyloid and tau pathology, therefore, enable clinicians to know whether AD pathology is present and may be causing or contributing to cognitive impairment. A more accurate diagnosis may enable better-informed clinician, patient, and caregiver decisions.

However, timely and accurate clinical diagnosis of dementia remains challenging, especially in its early stages when interventions could be most effective.^{10–12} Current diagnostic methods are typically based on clinical evaluation and structural neuroimaging, which may lack sensitivity and specificity, especially in the early stages of the disease.^{10,13,14} Cerebrospinal fluid (CSF) biomarkers and amyloid positron emission tomography (PET) are available that can detect AD brain pathology and thereby assist clinicians in determining whether AD is a possible cause of cognitive impairment. However, the use of CSF biomarkers and amyloid PET in dementia diagnosis has been limited for multiple reasons.¹⁵ These modalities are restricted to tertiary medical centers, which only a small fraction of the US population uses. CSF collection via lumbar puncture (LP) is often perceived as invasive and risky, requires an experienced LP team, and is contraindicated in some patients, such as those on anticoagulation. PET imaging can be costly, which can be prohibitive for widespread use, particularly in resource-limited settings. PET scans also subject patients to radiation exposure. Overall, the drawbacks of CSF biomarkers and PET have greatly limited their use in clinical dementia diagnosis.

In recent years, significant progress has been made in developing blood tests for AD. These blood tests include analytes measured in the blood serum or, more commonly, the blood plasma; the treatment of the blood sample determines whether serum or plasma is obtained. Biomarkers of amyloid and tau pathology include amyloid- β (A β), especially the A β 42 to A β 40 ratio and tau phosphorylated at positions 181, 217, and 231 (p-tau181, p-tau217, and p-tau231). It is important to note that while p-tau biomarkers have traditionally been thought of as representing tau neurofibrillary tangles, they typically have even more robust associations with amyloid pathology and may represent a response to amyloid pathology. Neurofilament light chain (NfL) is a biomarker of neuroaxonal injury and neurodegeneration, and glial fibrillary acidic protein (GFAP) is a biomarker of astrocyte activation.

Blood tests offer advantages over traditional CSF and imaging biomarkers, such as relative non-invasiveness, accessibility, and cost-effectiveness.^{10,16,17} Blood tests are now widely used in research studies and clinical trials and are becoming more available for clinical diagnosis.^{18–21} Integrating AD blood tests into clinical practice can revolutionize AD diagnosis and management by significantly reducing the time and cost involved in determining whether patients have AD pathology and may be eligible for AD-specific treatments and clinical trials of experimental anti-AD.^{22,23} Moreover, blood tests have the potential to provide an earlier and more accurate diagnosis of dementia, allowing for personalized approaches to care and resource allocation.^{17,24,25}

This review examines the current research status concerning AD blood tests, emphasizing their diagnostic, prognostic, and disease progression monitoring capabilities. With a specific focus on developments within the last 5 years, recent advancements in blood tests can potentially improve early detection, intervention, and research in AD, potentially revolutionizing dementia research and care.

2 | METHODS

We conducted a comprehensive search of databases including PubMed, Embase, and Google Scholar using specific search terms: “Alzheimer disease,” “Alzheimer disease and related dementia,” AND “plasma,” OR “blood” OR “serum” AND “biomarkers.” Additionally, we reviewed reference lists to identify relevant articles not captured in the initial search. Our inclusion criteria prioritized articles published in English and focused on plasma biomarkers in AD in the last 5 years.

3 | DISCUSSION

3.1 | CSF biomarkers of AD

CSF and imaging biomarkers have been instrumental in understanding AD pathophysiology. Breakthroughs in detecting AD pathology via CSF biomarkers and imaging modalities represented significant milestones in the field of Alzheimer research.^{25,26} The use of CSF biomarkers for AD gained prominence with the work of Trey Sunderland and colleagues. They found that CSF concentrations of A β 42 and tau were associated with AD pathology.^{27–31} These CSF biomarkers have become essential for diagnosing AD in research and clinical settings.

Initially, CSF biomarkers for AD, such as A β 42, total tau, and p-tau181, were primarily used in research settings.^{32,33} However, with advancements in technology and a growing understanding of AD pathology, several companies developed assays for detecting these biomarkers in CSF. There are now three assays for AD pathology that are approved by the United States Food and Drug Administration (FDA): the Lumipulse CSF A β 42/A β 40 test, and the Roche Elecsys p-tau181/A β 42 and total tau181/A β 42 tests.^{34–37} Appropriate use recommendations for CSF biomarkers have been described.³⁸

3.2 | PET biomarkers of AD

In 2004, William Klunk and Chester Mathis developed a groundbreaking imaging technique using a thioflavin derivative termed Pittsburgh Compound-B (PiB) as a PET tracer.^{39–41} PiB binds to amyloid plaques in the brain, allowing researchers to visualize and quantify the extent of amyloid deposition.⁴⁰

The FDA has approved various radioligands for clinical use, providing clinicians with valuable tools for detecting amyloid pathology in living individuals. The US FDA and the European Medicines Agency (EMA) have approved three radioligands, 18F-florbetapir, 18F-flutemetamol, and 18F-florbetaben, for amyloid PET imaging, enabling visualization of amyloid plaques in cognitively impaired individuals evaluated for AD and other causes of cognitive decline.^{42–45}

3.3 | Blood tests of AD

The FDA has not yet approved any blood tests for AD, but an increasing number are clinically available as laboratory-developed tests. Importantly, there is a wide variation in the performance and validation of these tests. The first clinically available test for AD was PrecivityAD, offered by C2N Diagnostics in St. Louis, Missouri, USA, which used plasma A β 42/A β 40, age, and apolipoprotein E prototype to provide a risk score for amyloid PET positivity.^{46–48} Additional blood tests for AD include a plasma A β 42/A β 40 test offered by Quest Diagnostics and a plasma A β 42/A β 40, p-tau181, and NfL test offered by Labcorp. More recently, the next generation of more accurate AD blood test based on p-tau217 have become available. The PrecivityAD2[®] blood test offered by C2N Diagnostics uses plasma A β 42/A β 40 and the ratio of phosphorylated to non-phosphorylated tau at position 217 to provide a likelihood of amyloid PET positivity.⁴⁹ A plasma p-tau217 test by ALZpath is also clinically available.⁵⁰

3.4 | Unique benefits of blood tests

Blood tests for AD offer several unique benefits compared to CSF analysis or PET imaging. The key advantages include accessibility, acceptability, cost-effectiveness, repetitive monitoring, and potential for population screening.^{16,21,51} Blood samples can be easily collected in a clinical setting or at home, making it convenient for patients, especially those in remote areas or with limited mobility. This accessibility allows for broader screening and early AD detection, leading to timely interventions and improved outcomes. Blood tests are generally more acceptable to patients than LP, which is required for CSF collection. The non-invasive nature of blood tests reduces patient discomfort and anxiety, leading to greater acceptance. This is particularly beneficial for older adults or individuals with cognitive impairment who may have difficulty with more complex procedures. In addition, blood tests for AD have the potential to be more cost-effective compared to CSF analysis or imaging tests.⁵² The more straightforward sample collection process and lower infrastructure requirements

reduce procedural costs. Additionally, the widespread availability of blood testing facilities further contributes to cost savings. As a result, blood tests for AD may offer a more affordable option for screening and diagnosing the disease.^{23,51} Additionally, blood tests can be performed multiple times without substantial difficulty, allowing for repetitive monitoring of AD biomarkers over time. This longitudinal assessment enables clinicians to track disease progression, evaluate treatment efficacy, and adjust management strategies. Regular monitoring of blood tests may also facilitate early detection of changes associated with AD pathology, enabling timely interventions to slow disease progression.⁵³

A significant advantage of blood tests for AD is their potential for widespread population screening.⁵⁴ This accessibility could facilitate early detection and intervention strategies for AD. This research underscores the potential for these biomarkers to be used in blood tests, reflecting treatment response and disease progression. Blood tests offer a promising solution for population-wide screening initiatives aimed at identifying individuals at risk of AD due to their accessibility, acceptability, and cost-effectiveness. Routine screening using blood tests could potentially enable early intervention and preventive measures, potentially reducing the burden of AD. This could potentially enhance patient outcomes and alleviate the overall burden of AD in the broader population.

3.5 | Potential challenges of blood tests

While blood tests for AD offer a non-invasive and accessible diagnostic option, their reliability and accuracy are under scrutiny, with varying performance levels across tests.⁵⁵ Factors such as age, sex, genetics, comorbidities, medications, and race/ethnicity can influence blood biomarker levels, potentially affecting diagnostic accuracy.^{55–61} A study by researchers at Washington University School of Medicine in St. Louis has shown that the race of people given AD blood tests may affect the interpretation of results.⁶² Three experimental blood tests used to identify people in the early stages of AD performed differently in Black individuals compared to White individuals, underscoring the need for reliable and accessible blood tests for AD. Additionally, in the United States, the costs of AD blood tests currently are not covered by insurance or reimbursed. Another significant limitation of the status of blood biomarkers is the lack of diversity in the research populations in which they have been evaluated, as the data have been derived from studies involving primarily White participants. This lack of representation means we know very little about the performance and reliability of these biomarkers in underrepresented groups (URGs). Addressing the variability and mitigating influencing factors in all populations will be essential for their widespread adoption and integration into clinical practice.^{56,57} Challenges remain in translating blood tests from research settings to routine clinical practice. Standardization of assay methods, validation across diverse populations, and integration into existing diagnostic algorithms are critical steps in realizing the full potential of plasma biomarkers for AD.

3.6 | Latest advancement in AD blood tests: State of the field

The field of blood tests for AD is rapidly evolving, with significant progress being made in recent years. Over the last 5 years, significant advancements have been made in developing blood tests for AD.^{6,21,50} Technological advancements have played a crucial role in developing plasma biomarkers for AD. Recent advancements in analytical techniques, including mass spectrometry, immunoassays, and multiplex assays, facilitate sensitive and accurate quantification of biomarkers in blood samples, thereby improving the detection of subtle changes linked to AD pathology.^{63–66} Machine learning and artificial intelligence (AI) techniques have been increasingly used to analyze complex plasma biomarker study datasets.^{66–68} These methods help identify patterns, biomarker combinations, and predictive models that can improve the accuracy of AD diagnosis and prognosis based on blood test profiles.⁶⁸ Machine learning algorithms can also aid in identifying individuals at risk of developing AD before clinical symptoms manifest.^{69,70}

Blood tests have emerged as valuable surrogate endpoints in clinical trials for evaluating the efficacy of potential AD therapeutics.^{71,72} For example, a study by Salloway et al.⁷³ highlighted the impact of gantenerumab and solanezumab on amyloid plaques using biomarkers. Additionally, a recent study by Wagemann et al.⁷⁴ underscored the potential of blood tests to reflect changes in brain pathology and treatment responses in AD. The study highlighted critical vital blood biomarkers, such as GFAP and neurogranin, which are crucial for non-invasive diagnostics. Observations included changes in GFAP and YKL-40 levels in response to gantenerumab treatment, indicating these markers' utility in monitoring therapeutic responses. This support using blood tests as cost-effective and accessible alternatives to CSF tests and neuroimaging, paving the way for personalized treatment strategies and improved patient care.

AD blood tests offer the potential for non-invasive, cost-effective, and easily accessible methods for diagnosing and monitoring AD pathology.¹⁷ Large-scale validation studies help establish blood test's reliability, sensitivity, and specificity across diverse populations. Regulatory agencies, including the FDA, are increasingly interested in validating blood tests for AD diagnosis and patient stratification in clinical settings.⁷⁵ Achieving regulatory approval requires rigorous validation and standardization of assays to ensure consistency and reproducibility.⁵

3.7 | Context of use for AD blood tests

3.7.1 | Research studies

AD blood tests are becoming integral to various research and clinical endeavors, from understanding disease mechanisms to diagnosis, monitoring, and treatment. They offer insights into molecular changes associated with AD, aiding in early detection, risk assessment, and differentiation from other dementias.¹⁷ Longitudinal studies employing

these biomarkers allow for disease monitoring and evaluation of intervention efficacy. Furthermore, they are pivotal in participant screening to determine eligibility for AD clinical trials and to inform population-based studies, contributing to public health strategies. Additionally, blood tests may guide healthcare decisions, optimizing resource allocation and improving healthcare efficiency. Blood tests' versatile applications underscore the importance of ongoing research and innovation to fully harness their potential in enhancing outcomes for individuals with AD and their families.

3.7.2 | Clinical trials

Clinical trials leverage blood tests of AD for participant selection, treatment effect assessment, response prediction, and complication identification. Biomarker screening ensures appropriate participant selection by detecting AD pathology, aiding in selecting individuals for symptomatic and prevention trials. For instance, elevated amyloid-beta and tau levels indicate AD pathology in those with cognitive impairment, guiding trial inclusion. In prevention trials targeting cognitively intact individuals, biomarkers identify those at risk of future AD, facilitating interventions to delay onset. Blood tests evaluate treatment effects on disease pathology and levels, supplementing clinical outcomes and imaging data.⁷⁶ They predict treatment response based on baseline levels or changes during trials, aiding treatment stratification.⁷⁷ Additionally, blood tests forecast complications, enabling proactive intervention.⁷⁸ Blood tests streamline participant selection, treatment assessment, response prediction, and complication identification in AD clinical trials, enhancing trial efficacy and intervention development.^{17,79}

3.7.3 | Clinical practice

Blood tests for AD have the potential to revolutionize diagnostic approaches and enhance patient care. Offering a non-invasive alternative to CSF analysis or PET scans, AD blood tests aid in diagnosing AD in symptomatic patients with cognitive impairment and treatment decisions.^{10,23} Primary care physicians (PCPs) or geriatricians, who may not routinely perform CSF analysis or PET scans, can use blood tests to screen patients for further evaluation by dementia specialists, optimizing resource allocation and ensuring timely referrals. Dementia specialists, like neurologists, leverage blood tests to diagnose AD and other neurodegenerative disorders, enhancing diagnostic accuracy and treatment selection. Biomarker-based algorithms identify patients likely to benefit from available treatments. Biomarker assessments, such as measuring p-tau217 or NfL, monitor disease progression and treatment response, guiding treatment adjustments.¹¹ In the future, blood tests may screen asymptomatic individuals for AD, enabling early interventions to delay or prevent dementia onset, thus reducing disease burden and improving long-term outcomes. Thus, blood tests offer valuable diagnostic and prognostic insights in clinical practice, fostering early detection, accurate diagnosis, and personalized

management of AD, potentially transforming AD diagnosis, treatment, and prevention.²³

3.7.4 | Future directions and conclusion

Despite the remarkable recent progress in blood tests for AD, several avenues for future research and development exist. These include refining assay techniques, longitudinal validation studies, and exploring novel biomarkers and multi-modal approaches. Ultimately, integrating blood biomarkers into routine clinical practice holds immense potential for improving early detection, prognostication, and therapeutic interventions in AD/ADRD. The emergence of these biomarkers represents a paradigm shift in AD/ADRD research, offering new insights into disease pathogenesis and progression. While challenges remain, ongoing efforts to harness blood tests' diagnostic and prognostic potential hold promise for transforming the landscape of AD/ADRD diagnosis and treatment.

CONFLICT OF INTEREST STATEMENT

Suzanne E Schindler has analyzed data provided by C2N Diagnostics to Washington University. She has served on a Scientific Advisory Board for Eisai and received speaker fees from Eli Lilly. John C Morris is funded by NIH grants #P30AG066444; P01AG003991; P01AG026276; Neither Dr. Morris nor his family owns stock or has equity interest (outside of mutual funds or other externally directed accounts) in any pharmaceutical or biotechnology company.

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REFERENCES

- Global status report on the public health response to dementia [Internet]. 2021 World Health Organization (WHO). Available from: <https://www.who.int/publications/i/item/9789240033245>
- Morris JC. The nosology of dementia. *Neurol Clin*. 2000;18(4):773–88.
- Morris JC. Dementia update 2003. *Alzheimer Dis Assoc Disord*. 2003;17(4):245–58.
- 2021 Alzheimer's disease facts and figures. *Alzheimers Dement*. 2021;17(3):327–406.
- Jack CR Jr, Bennett DA, Blennow K, Carrillo MC, Dunn B, Haeberlein SB, et al. NIA-AA research framework: toward a biological definition of Alzheimer's disease. *Alzheimers Dement*. 2018;14(4):535–62.
- Schindler SE. Fluid biomarkers in dementia diagnosis. *Continuum (Minneapolis)*. 2022;28(3):822–33.
- Tahami Monfared AA, Byrnes MJ, White LA, Zhang Q. The humanistic and economic burden of Alzheimer's disease. *Neurol Ther*. 2022;11(2):525–51.
- Bermejo-Pareja F, Del Ser T. Controversial past, splendid present, unpredictable future: a brief review of Alzheimer disease history. *J Clin Med*. 2024;13(2):536.
- Blennow K, de Leon MJ, Zetterberg H. Alzheimer's disease. *Lancet*. 2006;368(9533):387–403.
- Angioni D, Delrieu J, Hansson O, Fillit H, Aisen P, Cummings J, et al. Blood biomarkers from research use to clinical practice: what must be done? A report from the EU/US CTAD task force. *J Prev Alzheimers Dis*. 2022;9(4):569–79.
- Dokholyan NV, Mohs RC, Bateman RJ. Challenges and progress in research, diagnostics, and therapeutics in Alzheimer's disease and related dementias. *Alzheimers Dement (N Y)*. 2022;8(1):e12330.
- Pais M, Martinez L, Ribeiro O, Loureiro J, Fernandez R, Valiengo L, et al. Early diagnosis and treatment of Alzheimer's disease: new definitions and challenges. *Braz J Psychiatry*. 2020;42(4):431–41.
- Mueller SG, Weiner MW, Thal LJ, Petersen RC, Jack CR, Jagust W, et al. Ways toward an early diagnosis in Alzheimer's disease: the Alzheimer's disease neuroimaging initiative (ADNI). *Alzheimers Dement*. 2005;1(1):55–66.
- Gaugler JE, Kane RL, Johnston JA, Sarsour K. Sensitivity and specificity of diagnostic accuracy in Alzheimer's disease: a synthesis of existing evidence. *Am J Alzheimers Dis Other Dement*. 2013;28(4):337–47.
- Bouwman FH, Frisoni GB, Johnson SC, Chen X, Engelborghs S, Ikeuchi T, et al. Clinical application of CSF biomarkers for Alzheimer's disease: from rationale to ratios. *Alzheimers Dement (Amst)*. 2022;14(1):e12314.
- Balogun WG, Zetterberg H, Blennow K, Karikari TK. Plasma biomarkers for neurodegenerative disorders: ready for prime time? *Curr Opin Psychiatry*. 2023;36(2):112–8.
- Pais MV, Forlenza OV, Diniz BS. Plasma biomarkers of Alzheimer's disease: a review of available assays, recent developments, and implications for clinical practice. *J Alzheimers Dis Rep*. 2023;7(1):355–80.
- Hansson O. Biomarkers for neurodegenerative diseases. *Nat Med*. 2021;27(6):954–63.
- Schindler SE, Bateman RJ. Combining blood-based biomarkers to predict risk for Alzheimer's disease dementia. *Nat Aging*. 2021;1(1):26–8.
- Schindler SE, Barthélemy NR, Saef BA, Henson RL, Li Y, Benzinger TLS, et al. Blood-based biomarkers of Alzheimer disease as measured by a mass spectrometry assay are as accurate as CSF tests. *Alzheimers Dement*. 2023;19(S15):e070428.
- Hampel H, Hu Y, Cummings J, Mattke S, Iwatsubo T, Nakamura A, et al. Blood-based biomarkers for Alzheimer's disease: current state and future use in a transformed global healthcare landscape. *Neuron*. 2023;111(18):2781–99.
- Li Y, Schindler SE, Bollinger JG, Ovod V, Mawuenyega KG, Weiner MW, et al. Validation of plasma amyloid- β 42/40 for detecting Alzheimer disease amyloid plaques. *Neurology*. 2022;98(7):e688–99.
- Barthélemy NR, Salvadó G, Schindler SE, He Y, Janelidze S, Collij LE, et al. Highly accurate blood test for Alzheimer's disease is similar or superior to clinical cerebrospinal fluid tests. *Nat Med*. 2024;30(4):1085–95.
- Ferretti G, Serafini S, Angiolillo A, Monterosso P, Di Costanzo A, Matrone C. Advances in peripheral blood biomarkers of patients with Alzheimer's disease: moving closer to personalized therapies. *Biomed Pharmacother*. 2023;165:115094.
- Veitch DP, Weiner MW, Aisen PS, Beckett LA, DeCarli C, Green RC, et al. Using the Alzheimer's disease neuroimaging initiative to improve early detection, diagnosis, and treatment of Alzheimer's disease. *Alzheimers Dement*. 2022;18(4):824–57.
- Lee BC, Mintun M, Buckner RL, Morris JC. Imaging of Alzheimer's disease. *J Neuroimaging*. 2003;13(3):199–214.
- Sunderland T, Linker G, Mirza N, Putnam KT, Friedman DL, Kimmel LH, et al. Decreased beta-amyloid1-42 and increased tau levels in cerebrospinal fluid of patients with Alzheimer disease. *Jama*. 2003;289(16):2094–103.
- Sunderland T, Mirza N, Putnam KT, Linker G, Bhupali D, Durham R, et al. Cerebrospinal fluid beta-amyloid1-42 and tau in control subjects at risk for Alzheimer's disease: the effect of APOE epsilon4 allele. *Biol Psychiatry*. 2004;56(9):670–6.
- Sunderland T, Gur RE, Arnold SE. The use of biomarkers in the elderly: current and future challenges. *Biol Psychiatry*. 2005;58(4):272–6.

30. Sunderland T. Alzheimer disease. *South Med J*. 2005;98(6):588–9.
31. Sunderland T, Hampel H, Takeda M, Putnam KT, Cohen RM. Biomarkers in the diagnosis of Alzheimer's disease: are we ready? *J Geriatr Psychiatry Neurol*. 2006;19(3):172–9.
32. Holtzman DM. CSF biomarkers for Alzheimer's disease: current utility and potential future use. *Neurobiol Aging*. 2011;32(Suppl 1):S4–9.
33. Babić M, Svob Štrac D, Mück-Šeler D, Pivac N, Stanić G, Hof PR, et al. Update on the core and developing cerebrospinal fluid biomarkers for Alzheimer disease. *Croat Med J*. 2014;55(4):347–65.
34. Schindler SE, Gray JD, Gordon BA, Xiong C, Batrla-Utermann R, Quan M, et al. Cerebrospinal fluid biomarkers measured by Elecsys assays compared to amyloid imaging. *Alzheimers Dement*. 2018;14(11):1460–9.
35. Hansson O, Seibyl J, Stomrud E, Zetterberg H, Trojanowski JQ, Bittner T, et al. CSF biomarkers of Alzheimer's disease concord with amyloid- β PET and predict clinical progression: a study of fully automated immunoassays in BioFINDER and ADNI cohorts. *Alzheimers Dement*. 2018;14(11):1470–81.
36. Kaplow J, Vandijck M, Gray J, Kanekiyo M, Huyck E, Traynham CJ, et al. Concordance of Lumipulse cerebrospinal fluid t-tau/A β 42 ratio with amyloid PET status. *Alzheimers Dement*. 2020;16(1):144–52.
37. Gobom J, Parnetti L, Rosa-Neto P, Vyhalek M, Gauthier S, Cataldi S, et al. Validation of the LUMIPULSE automated immunoassay for the measurement of core AD biomarkers in cerebrospinal fluid. *Clin Chem Lab Med*. 2021;60(2):207–19.
38. Shaw LM, Arias J, Blennow K, Galasko D, Molinuevo JL, Salloway S, et al. Appropriate use criteria for lumbar puncture and cerebrospinal fluid testing in the diagnosis of Alzheimer's disease. *Alzheimers Dement*. 2018;14(11):1505–21.
39. Mathis CA, Wang Y, Klunk WE. Imaging beta-amyloid plaques and neurofibrillary tangles in the aging human brain. *Curr Pharm des*. 2004;10(13):1469–92.
40. Klunk WE, Engler H, Nordberg A, Wang Y, Blomqvist G, Holt DP, et al. Imaging brain amyloid in Alzheimer's disease with Pittsburgh compound-B. *Ann Neurol*. 2004;55(3):306–19.
41. Wang Y, Klunk WE, Debnath ML, Huang GF, Holt DP, Shao L, et al. Development of a PET/SPECT agent for amyloid imaging in Alzheimer's disease. *J Mol Neurosci*. 2004;24(1):55–62.
42. Hampel H, Au R, Mattke S, van der Flier WM, Aisen P, Apostolova L, et al. Designing the next-generation clinical care pathway for Alzheimer's disease. *Nat Aging*. 2022;2(8):692–703.
43. US Food and Drug Administration. Amyvid (florbetapir F18 injection) [highlights of prescribing information]. 2012 https://www.accessdata.fda.gov/drugsatfda_docs/label/2012/202008s000lbl.pdf
44. US Food and Drug Administration. Vizamyl (flutemetamol F18 injection) [highlights of prescribing information]. 2013 https://www.accessdata.fda.gov/drugsatfda_docs/label/2013/203137s000lbl.pdf
45. US Food and Drug Administration. Neuraceq (florbetaben F18 injection) [highlights of prescribing information]. 2014 https://www.accessdata.fda.gov/drugsatfda_docs/label/2014/204677s000lbl.pdf
46. West T, Kirmess KM, Meyer MR, Holubasch MS, Knapik SS, Hu Y, et al. A blood-based diagnostic test incorporating plasma A β 42/40 ratio, ApoE genotype, and age accurately identifies brain amyloid status: findings from a multi cohort validity analysis. *Mol Neurodegener*. 2021;16(1):30.
47. Schindler SE, Bollinger JG, Ovod V, Mawuenyega KG, Li Y, Gordon BA, et al. High-precision plasma β -amyloid 42/40 predicts current and future brain amyloidosis. *Neurology*. 2019;93(17):e1647–59.
48. Hu Y, Kirmess KM, Meyer MR, Rabinovici GD, Gatsonis C, Siegel BA, et al. Assessment of a plasma amyloid probability score to estimate amyloid positron emission tomography findings among adults with cognitive impairment. *JAMA Netw Open*. 2022;5(4):e228392.
49. Meyer MR, Kirmess KM, Eastwood S, Wenthe-Roth TL, Irvin F, Holubasch MS, et al. Clinical validation of the PrecivityAD2 blood test: a mass spectrometry-based test with algorithm combining %p-tau217 and A β 42/40 ratio to identify presence of brain amyloid. *Alzheimers Dement*. 2024;20(5):3179–92.
50. Ashton NJ, Brum WS, Di Molfetta G, Benedet AL, Arslan B, Jonaitis E, et al. Diagnostic accuracy of a plasma phosphorylated tau 217 immunoassay for Alzheimer disease pathology. *JAMA Neurol*. 2024;81(3):255–63.
51. Hansson O, Blennow K, Zetterberg H, Dage J. Blood biomarkers for Alzheimer's disease in clinical practice and trials. *Nat Aging*. 2023;3(5):506–19.
52. Noda K, Lim Y, Goto R, Sengoku S, Kodama K. Cost-effectiveness comparison between blood biomarkers and conventional tests in Alzheimer's disease diagnosis. *Drug Discov Today*. 2024;29(3):103911.
53. Counts SE, Ikonovic MD, Mercado N, Vega IE, Mufson EJ. Biomarkers for the early detection and progression of Alzheimer's disease. *Neurotherapeutics*. 2017;14(1):35–53.
54. Leuzy A, Mattsson-Carlén N, Palmqvist S, Janelidze S, Dage JL, Hansson O. Blood-based biomarkers for Alzheimer's disease. *EMBO mol Med*. 2022;14(1):e14408.
55. Zetterberg H, Burnham SC. Blood-based molecular biomarkers for Alzheimer's disease. *Mol Brain*. 2019;12(1):26.
56. Schindler SE, Karikari TK. Comorbidities confound Alzheimer's blood tests. *Nat Med*. 2022;28(7):1349–51.
57. Mielke MM, Dage JL, Frank RD, Algeciras-Schimmich A, Knopman DS, Lowe VJ, et al. Performance of plasma phosphorylated tau 181 and 217 in the community. *Nat Med*. 2022;28(7):1398–405.
58. Doecke JD, Laws SM, Faux NG, Wilson W, Burnham SC, Lam CP, et al. Blood-based protein biomarkers for diagnosis of Alzheimer disease. *Arch Neurol*. 2012;69(10):1318–25.
59. Kiddle SJ, Thambisetty M, Simmons A, Riddoch-Contreras J, Hye A, Westman E, et al. Plasma based markers of [11C] PiB-PET brain amyloid burden. *PLoS One*. 2012;7(9):e44260.
60. O'Bryant SE, Xiao G, Barber R, Reisch J, Doody R, Fairchild T, et al. A serum protein-based algorithm for the detection of Alzheimer disease. *Arch Neurol*. 2010;67(9):1077–81.
61. Mattsson N, Zetterberg H, Janelidze S, Insel PS, Andreasson U, Stomrud E, et al. Plasma tau in Alzheimer disease. *Neurology*. 2016;87(17):1827–35.
62. Schindler SE, Karikari TK, Ashton NJ, Henson RL, Yarasheski KE, West T, et al. Effect of race on prediction of brain amyloidosis by plasma A β 42/A β 40, phosphorylated tau, and neurofilament light. *Neurology*. 2022;99(3):e245–57.
63. O'Bryant SE, Gupta V, Henriksen K, Edwards M, Jeromin A, Lista S, et al. Guidelines for the standardization of preanalytic variables for blood-based biomarker studies in Alzheimer's disease research. *Alzheimers Dement*. 2015;11(5):549–60.
64. Janelidze S, Stomrud E, Palmqvist S, Zetterberg H, van Westen D, Jeromin A, et al. Plasma β -amyloid in Alzheimer's disease and vascular disease. *Sci Rep*. 2016;6:26801.
65. Lista S, Toschi N, Baldacci F, Zetterberg H, Blennow K, Kilimann I, et al. Diagnostic accuracy of CSF neurofilament light chain protein in the biomarker-guided classification system for Alzheimer's disease. *Neurochem Int*. 2017;108:355–60.
66. Verberk IMW, Slot RE, Verfaillie SCJ, Heijst H, Prins ND, van Berckel BNM, et al. Plasma amyloid as Prescreeener for the earliest Alzheimer pathological changes. *Ann Neurol*. 2018;84(5):648–58.
67. Chang CH, Lin CH, Lane HY. Machine learning and novel biomarkers for the diagnosis of Alzheimer's disease. *Int J Mol Sci*. 2021;22(5):2761.
68. Winchester LM, Harshfield EL, Shi L, Badhwar A, Khleifat AA, Clarke N, et al. Artificial intelligence for biomarker discovery in Alzheimer's disease and dementia. *Alzheimers Dement*. 2023;19(12):5860–71.
69. Kumar S, Oh I, Schindler S, Lai AM, Payne PRO, Gupta A. Machine learning for modeling the progression of Alzheimer disease dementia

- using clinical data: a systematic literature review. *JAMIA Open*. 2021; 4(3):ooab052.
70. Yue L, Chen WG, Liu SC, Chen SB, Xiao SF. An explainable machine learning based prediction model for Alzheimer's disease in China longitudinal aging study. *Front Aging Neurosci*. 2023;15:1267020.
 71. Hampel H, Cummings J, Blennow K, Gao P, Jack CR Jr, Vergallo A. Developing the ATX(N) classification for use across the Alzheimer disease continuum. *Nat Rev Neurol*. 2021;17(9):580–9.
 72. Hansson O, Edelmayer RM, Boxer AL, Carrillo MC, Mielke MM, Rabinovici GD, et al. The Alzheimer's Association appropriate use recommendations for blood biomarkers in Alzheimer's disease. *Alzheimers Dement*. 2022;18(12):2669–86.
 73. Salloway S, Farlow M, McDade E, Clifford DB, Wang G, Llibre-Guerra JJ, et al. A trial of gantenerumab or solanezumab in dominantly inherited Alzheimer's disease. *Nat Med*. 2021;27(7):1187–96.
 74. Wagemann O, Liu H, Wang G, Shi X, Bittner T, Scelsi MA, et al. Downstream biomarker effects of gantenerumab or Solanezumab in dominantly inherited Alzheimer disease: the DIAN-TU-001 randomized clinical trial. *JAMA Neurol*. 2024;29:585–90.
 75. Frisoni GB, Molinuevo JL, Altomare D, Carrera E, Barkhof F, Barkhof J, et al. Precision prevention of Alzheimer's and other dementias: anticipating future needs in the control of risk factors and implementation of disease-modifying therapies. *Alzheimers Dement*. 2020;16(10):1457–68.
 76. Bermudez C, Graff-Radford J, Syrjanen JA, Stricker NH, Algeciras-Schimnich A, Kouri N, et al. Plasma biomarkers for prediction of Alzheimer's disease neuropathologic change. *Acta Neuropathol*. 2023; 146(1):13–29.
 77. Cummings J. The role of biomarkers in Alzheimer's disease drug development. *Adv Exp Med Biol*. 2019;1118:29–61.
 78. Zhang F, Petersen M, Johnson L, Hall J, O'Bryant SE. Combination of serum and plasma biomarkers could improve prediction performance for Alzheimer's disease. *Genes (Basel)*. 2022;13(10): 1738.
 79. Aisen PS, Bateman RJ, Carrillo M, Doody R, Johnson K, Sims JR, et al. Platform trials to expedite drug development in Alzheimer's disease: a report from the EU/US CTAD task force. *J Prev Alzheimers Dis*. 2021;8(3):306–12.

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